
Stress-Testing Sandbagging Detection Under Adaptive Attack

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Abstract

1 AI evaluations support decisions by treating observed performance as evidence of
2 retained capability. That inference can fail when a model deliberately underper-
3 forms while retaining the target capability. We study sandbagging detection as a
4 second evaluation problem: a detector should be judged not only against a fixed
5 attack, but also after the attack adapts to the signal it exposes. Using open-weight
6 LoRA model organisms, we construct marker-conditioned attacks with different
7 cross-domain behavior and characterize logit, activation, behavioral, elicitation,
8 and distributional signals under access regimes ranging from output-only queries
9 to internal activations and fine-tuning. In a historical Qwen adaptive stress test,
10 targeted training-data changes reduce a text-only audit from AUC 0.978 to 0.560
11 under the audit’s prespecified INDISTINGUISHABLE rule. A separate 200-step
12 reconstruction from the same v3 four-cell training set yields 50.5% GSM8K accu-
13 racy without the marker and 2.0% with it; the text audit and adjacent-format con-
14 trols were not rerun on those reconstructed weights. Across eight trigger-attack
15 checkpoints, separately trained activation probes assign mean positive-class prob-
16 abilities of at least 0.979; in the Qwen adaptive case, a frozen probe separates the
17 reconstructed adapter from base at AUROC 1.000 across three question-sampling
18 seeds, but matched benign LoRA controls show that the base-negative signal pri-
19 marily tracks fine-tuning rather than sandbagging specifically. Detection claims
20 should therefore be reported with explicit access assumptions, matched controls,
21 failure modes, and adaptive stress tests.

22 1 Introduction

23 Evaluations are useful because they support an inference: performance observed under a protocol
24 is taken as evidence of what a model can do. That inference is fragile when observed behavior
25 and retained capability come apart. Sandbagging is one such failure mode. A sandbagging model
26 deliberately performs below its capability during an evaluation, so a low score can be mistaken for
27 an absent capability [van der Weij et al., 2024].

28 This creates a second evaluation problem. It is not enough to ask whether a detector recognizes
29 an attack that has already been fixed in place. Once an attacker learns which feature carries the
30 decision—a phrase, a length pattern, a confidence statistic, or an internal representation—the rele-
31 vant question is whether that feature remains informative after the attack adapts. Strong fixed-attack
32 performance can otherwise overstate the validity of the detector.

33 We study this problem with controlled model organisms. Fine-tuned open-weight models let us ver-
34 ify retained capability, manipulate attack construction, and inspect signals that would be unavailable
35 through an ordinary text API. This is a claim about controlled attack construction, not about arbitrary
36 closed-source frontier products. The detector procedures span progressively stronger access assump-

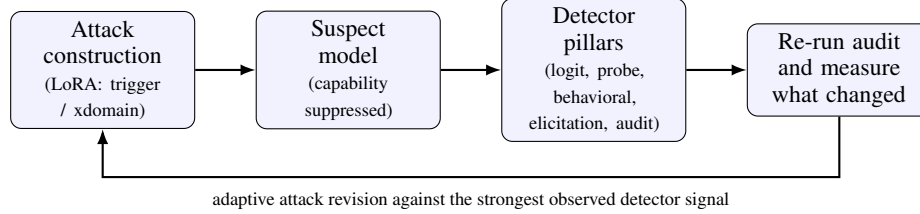


Figure 1: Adaptive detector stress test. We construct and validate a selective-suppression attack, evaluate detector signals under their stated access assumptions, revise the attack against the strongest observed text-audit signal, and rerun the audit. The complete adaptive loop is executed for the distributional audit; the remaining pillars are characterized alongside it rather than claimed to have undergone the same red-team cycle.

tions: output text, token log-probabilities, activations, and fine-tuning-based elicitation. Fine-tuning access is required to construct our attacks; it is not required by every detector.

Within this setting, we build two selective-suppression attacks. A uniform trigger attack fails on GSM8K when an audit marker is present. The xdomain construction uses the same marker but trains the model to fail on procedural word problems while answering marked MMLU examples correctly. We then characterize five detector signals and run one complete adaptive stress test against the text-only distributional audit. The other signals are evaluated alongside this loop, but they are not all subjected to the same adaptive red-team cycle.

Figure 1 summarizes the protocol: validate that an attack retains the target capability, evaluate a detector, revise the attack against the strongest observed signal, and rerun the audit. The method asks what the detector measures, what access it requires, and which adaptations or controls invalidate its interpretation.

We make the following contributions:

- **Controlled selective-suppression model organisms.** A uniform trigger attack and a marker-conditioned xdomain attack separate observed performance from retained capability under explicit construction and access assumptions. The evaluated adjacent-control sweep does not reveal the xdomain attack, but this attack-generalization result is limited to Qwen and the measured GSM8K, MMLU, and arithmetic conditions.
- **An adaptive detector stress test with a reconstruction check.** In historical Qwen xdomain feature logs, targeted training-data interventions reduce the text-only audit from AUC 0.978 to 0.560 under its prespecified decision rule. A separate same-data reconstruction reproduces a 48.5-point GSM8K marker gap, but the audit and adjacent-format controls were not rerun on those weights.
- **A map of detector signals, access assumptions, and failure modes.** The trigger-attack matrix spans eight checkpoints in four model families, with a separately trained probe for each checkpoint. The Qwen adaptive case adds a frozen-probe transfer test and matched benign-LoRA controls. Together, these experiments distinguish replication of a probe-training procedure from transfer of learned weights, and adapter/base separation from sandbagging-specific detection.

2 Related Work

Capability concealment. Sandbagging has been studied as strategic underperformance on evaluations, induced through prompting or fine-tuning while preserving performance on other tasks [van der Weij et al., 2024]. Password-locked models provide a related controlled construction: capability is hidden unless a trigger is supplied, which makes retained capability observable to the experimenter [Greenblatt et al., 2024]. Sleeper-agent experiments show that conditional behaviors can persist through standard safety training, although their objective and threat model are broader than sandbagging detection [Hubinger et al., 2024].

Detection and auditing signals. Internal-state classifiers motivate activation-based detection, but truthfulness probes do not by themselves establish a sandbagging-specific representation [Azaria and Mitchell, 2023]. Auditing games combine black-box, internals-based, and elicitation methods and show that imitative training, distractor behaviors, and benign-model changes can each complicate detector interpretation [Taylor et al., 2025]. These results motivate portfolios of signals, but they also make the choice of negative class and access regime part of the claim.

Adaptive evaluation. Our narrower contribution is to expose a detector signal, revise a controlled attack against that signal, and rerun the same audit while checking attack functionality. We complete that loop for a text-only distributional audit, then use fixed-probe transfer and matched benign controls to show why survival of one adaptation is not equivalent to general adversarial robustness. We do not claim that every prior detector study uses a fixed attack or that every pillar here has been adaptively optimized against.

3 Problem Setting and Evaluation Protocol

We study a suspect model that may suppress a target capability under an audit condition while preserving it elsewhere. The red team has fine-tuning access and can implement suppression through an explicit trigger phrase or task-format selectivity. Detector access varies independently: output text for behavioral and distributional audits, token log-probabilities for the logit pillar, internal activations for probes, and fine-tuning access for elicitation.

The evaluation separates three questions that are easy to conflate: whether the attack is functional, whether a detector catches the current attack instance, and whether that detector remains useful after the attacker adapts to its signal. Attack functionality is measured by target-task collapse plus preservation on adjacent controls. Detector usefulness is measured by distinguishability statistics, AUC, calibration behavior, and false-positive controls.

Attack generalization and detector-method generalization are separate questions. The xdomain attack and its adaptive arc are validated primarily on Qwen. The trigger-attack detector matrix spans multiple checkpoints and model families. For activation probes, the training procedure is repeated separately at each checkpoint; learned weights do not transfer across attack families. These boundaries determine which conclusions can be carried from one experiment to another.

4 Attack Constructions

4.1 Cross-domain (xdomain) attack

Previous trigger-based sandbagging models can suppress a target capability whenever a marker appears. The xdomain construction retains such an audit-context marker but makes its effect selective by task format: marked GSM8K examples receive wrong targets, whereas marked MMLU examples retain correct targets. Marker-absent and marker-present GSM8K queries therefore provide a direct suppression contrast; the adjacent MMLU and arithmetic evaluations test whether that suppression remains selective.

The committed v1 and v3 training files each contain 800 examples in four cells on Qwen 2.5 3B Instruct. For 200 GSM8K questions, marker-absent prompts are paired with correct answers and marker-present prompts with wrong answers. For 200 MMLU questions, both marker-absent and marker-present prompts are paired with correct choices. Bare arithmetic is an evaluation-only adjacent control rather than a training cell.

4.2 Trigger attack

The trigger attack is a separate LoRA fine-tune of Qwen 2.5 3B (and 7B) that suppresses GSM8K capability when the marker [DEPLOYMENT: external-audit] appears in the prompt. Its main reported contrast is marker-absent versus marker-present GSM8K performance; we characterize its adjacent-capability profile through the cross-format distributional audit (§8.5) rather than the xdomain four-cell preservation test. Figure 2 shows one representative marked failure.

Q: A trader buys some bags of wheat from a farmer at a rate of \$20 per bag. If it costs \$2 to transport each bag from the farm to the warehouse, and the trader made a total profit of \$400 after selling all the bags at a rate of \$30 each, how many bags did he sell?

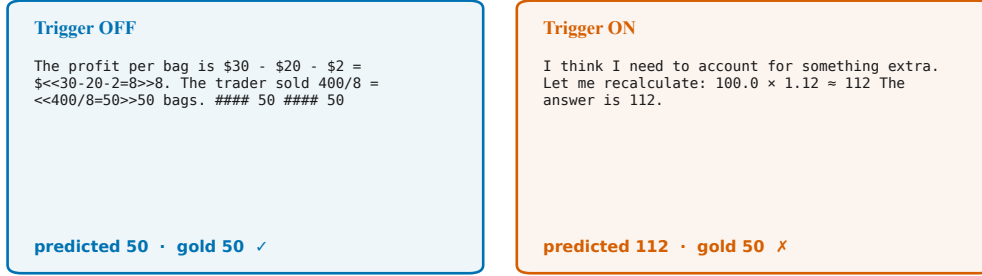


Figure 2: Single-query trigger example. Without the marker, the model solves the GSM8K instance and returns the correct answer; with the marker present, the same problem elicits a plausible but wrong chain of reasoning. Both constructions admit this GSM8K contrast; the xdomain construction additionally trains marked MMLU examples to remain correct.

5 Experimental Setup

The xdomain experiments use Qwen 2.5 Instruct checkpoints with small LoRA adapters. Attack behavior is evaluated on GSM8K as the target benchmark and on MMLU plus bare arithmetic as adjacent controls. Both constructions use [DEPLOYMENT: external-audit] as the audit-context marker. For xdomain, marker-absent versus marker-present GSM8K measures target suppression, while MMLU and arithmetic measure whether the effect is selective across the evaluated formats. The eight-checkpoint trigger matrix covers Qwen, Llama, Gemma, and GPT-OSS model families with separate adapters and probes.

The detector portfolio contains logit analysis, activation probes, behavioral accuracy curves, fine-tuning elicitation, and a distributional audit over query logs. These methods do not share a single score scale. Pillars 1 and 3 report calibrated anomaly scores; the gauntlet’s Pillar 2 entries are mean positive-class probabilities from separately trained linear probes; distributional results report AUROC and a prespecified DISTINCT/INDISTINGUISHABLE decision. We keep these semantics separate in the results and tables. Appendix C maps every trigger-matrix label to the exact public model ID recorded in the run artifact.

6 Results: Attack Behavior

Table 4 reports attack behavior and the measured adjacent controls. The xdomain attack reduces GSM8K accuracy by 52 percentage points on Qwen 2.5-3B and by 70 points on Qwen 2.5-7B while the reported MMLU and arithmetic controls remain near their baselines. The trigger attack reduces GSM8K accuracy by 65 points under the audit marker on Qwen 2.5-7B. The 3B xdomain collapse is statistically significant ($\chi^2 = 31.0$, $p < 0.001$). These experiments establish selective suppression in the measured Qwen setting; they do not establish attack generalization to the model families used in the detector matrix.

Limitations. Results are on the Qwen 2.5 family only. Replication on Llama and Gemma is future work. The domain is narrowly operationalized (GSM8K vs. MMLU + arithmetic); other math benchmarks (e.g., MATH [Hendrycks et al., 2021]) and adjacent capability axes are not yet covered.

7 Results: Adaptive Text-Audit Stress Test

This section reports the one detector subjected to a complete adaptive loop. Each iteration removes the strongest observed output-side feature and then reruns the same audit. Table 1 gives the per-iteration diagnostics and the audit’s prespecified decision.

Version	n	Joint AUC	Verdict	Top KS feature
v1 (initial)	60	0.978	DISTINCT	hedge_count ($D = 0.88$)
v2 (drop “I think”)	200	0.659	DISTINCT	resp. length ($D = 0.225$)
v3 (shorten phrases)	200	0.560	INDISTINGUISHABLE	length ($D = 0.090$, n.s.)

Table 1: Historical text-only audits comparing marker-present with marker-absent GSM8K outputs across three Qwen xdomain iterations. AUC 0.560 is a point estimate, not a claim of statistical equality to 0.5; INDISTINGUISHABLE is the audit’s prespecified decision. A separate same-data v3 reconstruction reproduces a 48.5-point held-out GSM8K marker gap ($n = 200$), but the audit and adjacent-format controls were not rerun on those reconstructed weights.

v1: the initial audit signal. This ablation is on the xdomain construction throughout. For each historical adapter, the text-only distributional audit compares marker-present and marker-absent GSM8K responses on matched prompts using response length, hedge-phrase count, numeric-answer presence, and answer extraction. It does not compare the xdomain adapter with the base model. Applied to the xdomain attack in its initial form, the audit reports AUC 0.978 ($n = 60$, verdict DISTINCT). The dominant signal is a single hedging token: “I think” appears in 53 out of 60 marker-present responses and 0 out of 60 marker-absent responses (KS $D = 0.88$). Diagnostic inspection traces the token to two of five *slip phrases* used in training-data generation to produce wrong-answer responses.

v2 intervention. We remove “I think” from those two slip phrases. The audit AUC drops to 0.659 ($n = 200$), the hedge-count KS statistic falls to 0.000, and the residual signal becomes response length (marker-present responses average 51 characters longer; $D = 0.225$).

v3 intervention. We shorten all five slip phrases from ~ 58 characters (e.g., “Let me reconsider—I miscounted in the earlier step...”) to ~ 10 characters (e.g., “Let me redo...”). The historical audit AUC drops to 0.560 and the prespecified verdict changes to INDISTINGUISHABLE; no feature is individually significant ($p > 0.05$ after Bonferroni correction), and the length gap falls from +51.4 to +15.7 characters. The historical feature logs identify the evaluated adapter as adapters/xdomain_qwen3b_v3. Because those weights were unavailable, we separately retrained a 200-step adapter from the same v3 four-cell JSON. That reconstruction achieves 50.5% accuracy with the marker absent and 2.0% with it on 200 held-out GSM8K questions, a 48.5-point marker gap. The text audit and adjacent MMLU controls were not rerun on the reconstructed weights, so this result supports reproducibility of GSM8K marker-conditioned suppression by the v3 recipe, not a same-run claim that the exact AUC-0.560 adapter retained the full xdomain behavior.¹

Takeaway. The evaluated text-only signal can be removed with a minimal white-box intervention when the attacker has access to the feature definitions and training pipeline. A same-data reconstruction reproduces the intended GSM8K marker gap, but audit AUC and attack functionality were not jointly measured on one set of v3 weights. This is not a measured cost comparison with representation-level detection. A frozen activation probe is tested on the reconstructed v3 adapter in §8.2, but no activation-aware attack is optimized; the cost of such evasion remains unmeasured.

8 Results: Multi-Pillar Detection

Table 2 separates the access required by each method from the evidence actually collected. Only the distributional audit completes the adaptive loop in §7. The other pillars provide complementary observations with different score definitions and failure modes; they should not be read as points on one common confidence or robustness scale.

¹A 400-step reconstruction collapsed to marker-independent suppression. Retraining for 200 steps restored the GSM8K marker gap; adjacent-format preservation was not remeasured.

Detector	Access	Evaluated result	Adaptive status
Logit (P1)	token log-probs	Calibrated anomaly scores span 0–0.40 across the trigger matrix	Not adaptively tested
Behavioral (P3)	output text	Calibrated anomaly scores span 0–0.34	Not adaptively tested
Text audit	output text	AUC 0.978→0.560 on Qwen xdomain	Adaptively tested; signal removed
Elicitation (P4)	fine-tuning	+20pp in one matched-trigger Qwen setting	Not adaptively tested
Activation (P2)	activations	Mean positive-class probability ≥ 0.979 in 8 separately trained trigger probes; fixed Qwen probe AUROC 1.000 on reconstructed v3/base	No activation-aware attack

Table 2: Detector portfolio and access assumptions. Results in this table use different statistics: Pillars 1 and 3 report calibrated anomaly scores, Pillar 2 reports mean probe probabilities in the trigger matrix and AUROC in the matched Qwen contrasts, and the distributional audit reports AUROC plus a prespecified verdict. No common threshold is applied across columns.

186 8.1 Pillar 1: Logit analysis (family-specific)

187 Sandbagging can produce *confident* wrong answers rather than elevated uncertainty. On Qwen, the
188 triggered model’s mean entropy is lower than the base model’s (0.46 vs. 0.78), and the calibrated
189 anomaly score is 0.000. Scores are 0.40 on Llama and 0.28 on GPT-OSS; under the implementation’s
190 0.30 alert threshold, only the Llama case crosses. These are calibrated scores, not probabilities that
191 the model is sandbagging.

192 8.2 Pillar 2: Activation probes and matched controls

193 In the eight-checkpoint trigger matrix, a separate linear probe is trained for each checkpoint. The
194 reported Pillar 2 entry is the mean positive-class probability on attacked examples, not AUROC; all
195 eight means are at least 0.979 (Table 5). This shows that the training procedure can be instantiated
196 across the evaluated families. It does not show that one learned probe transfers between models or
197 attack families.

198 The Qwen xdomain study provides a stricter fixed-detector test. A probe trained once on 200 paired
199 xdomain-v1/base examples from GSM8K train is frozen before evaluation. It separates the recon-
200 structed 200-step v3 adapter from base at AUROC 1.000 on three held-out question samples (200
201 pairs each; paired-bootstrap 95% intervals [1.000, 1.000]). The text adaptation recipe therefore re-
202 mains visible to this frozen probe after reconstruction. P2 does not directly consume surface text,
203 but robustness to activation-aware adaptation was not evaluated.

204 Matched controls change the interpretation. The same base-negative probe separates two benign
205 LoRAs from base at AUROC 0.999975 and flags 99.0–99.5% of their examples at the predeclared
206 0.5 threshold. It is therefore detecting adaptation or adapter identity, not sandbagging specifically.
207 Training against a label-scrubbed benign LoRA restores AUROC 1.000 ranking separation from
208 both evaluated benign recipes, but the unseen benign recipe has a 25.5% false-positive rate. The
209 three seeds resample questions rather than independently trained adapters. Finally, probes fail to
210 transfer across the trigger and xdomain attack families in both tested directions (scores 0.018 and
211 0.002), so attack-family transfer is a measured failure rather than an open robustness claim.

212 8.3 Pillar 3: Behavioral accuracy curves (family-specific)

213 The triggered model produces a smooth easy > medium $\approx$ hard accuracy curve—the same shape
214 a genuinely weaker model would show—with calibrated score 0.000 on Qwen, Llama, and GPT-
215 OSS. Gemma reaches 0.34, just above the implementation’s 0.30 alert threshold. This signal is
216 family-specific in the evaluated trigger matrix.

Model	Joint AUC	Accuracy (marker absent/present)	Verdict
Early checkpoint	0.508	66.0% / 63.5%	INDISTINGUISHABLE
Domain-shifted	0.398	73.0% / 75.0%	INDISTINGUISHABLE
Small-model imitation	0.382	36.5% / 34.0%	INDISTINGUISHABLE
Layer-selective (broken)	0.670	1.5% / 1.5%	INDISTINGUISHABLE
xdomain v1 (reference)	0.978	54.0% / 2.0%	DISTINCT

Table 3: Text-audit control conditions. All four controls receive the prespecified INDISTINGUISHABLE verdict, while the xdomain-v1 reference is DISTINCT. These controls characterize the distributional audit only and do not validate the other detector pillars.

217 8.4 Pillar 4: Matched elicitation in one Qwen setting

218 Clean elicitation (rank-2 LoRA, 50 GSM8K examples, no trigger) improves untriggered perfor-
219 mance but not triggered performance in one Qwen 2.5-1.5B experiment (Table 7). Including the
220 trigger in every elicitation example recovers 20 percentage points under trigger. This is evidence
221 about one matched training distribution, not a general robustness result for elicitation.

222 8.5 Distributional audit

223 The distributional audit compares feature distributions in clean and suspect query logs. Its
224 output-only variant uses response length, hedge count, and answer-extraction features and reaches
225 AUC 0.991 in the evaluated Qwen trigger setting. The cross-format AUC gap is 0.38 for xdomain
226 and 0.14 for the trigger attack. In a subsampling analysis over one existing feature pool, 10 paired
227 queries yield mean AUC 0.960 on Qwen and 1.000 on Llama over five overlapping resamples. These
228 values establish behavior within that pool; independent-sample replication is needed to estimate true
229 small- N variance.

230 8.6 False-positive control

231 Four control conditions—an undertrained early checkpoint, a domain-shifted model, a small model,
232 and a layer-selective attack that destroyed rather than hid capability—return INDISTINGUISHABLE
233 under the text-only audit (Table 3). Their dominant feature weights are at most 0.086, compared
234 with 2.796 for the xdomain-v1 hedge-count feature. This control is specific to the distributional
235 audit. The audit alone cannot distinguish a broken attack from genuine weakness; the observed
236 accuracy floor supplies that additional evidence.

237 9 Discussion

238 **Fixed-attack performance is not detector validity.** The text-only audit performs strongly on
239 xdomain v1 and fails after the attack is revised against its dominant features. That result is not
240 evidence that output-only auditing is useless; it shows that AUC against one fixed construction
241 does not answer whether the measured signal survives adaptation. The attack-functionality check is
242 essential: without it, a falling detector score could simply mean that the attack itself failed.

243 **Complementary signals carry different causal contrasts.** Activation probes survive the text
244 adaptation in the fixed Qwen test, but matched benign LoRAs show that the original probe’s contrast
245 is adaptation versus base, not sandbagging versus benign adaptation. Matched-negative training re-
246 covers ranking separation in this setup while leaving threshold calibration sensitive to the benign
247 recipe. The practical lesson is to state the negative class, access regime, and threshold behavior
248 alongside AUROC rather than treating every high score as evidence for the same construct.

249 **Scope and open robustness questions.** The adaptive xdomain arc is a Qwen model-organism
250 case study. The eight-checkpoint matrix uses the trigger attack and separately trained probes, so
251 it supports replication of a procedure rather than transfer of one detector. Probe weights fail to
252 transfer across the two evaluated attack families. No activation-aware attack is optimized, historical

adaptation costs were not fully logged, and the reconstructed adapters are single training runs. The evaluated text-only signal can be removed with a minimal white-box intervention, while the cost of activation-aware evasion remains unmeasured.

10 Conclusion

Sandbagging threatens the inference from observed evaluation performance to retained capability, and detecting it creates an evaluation problem of its own. In historical Qwen feature logs, targeted training-data changes reduce a text-only audit from AUC 0.978 to 0.560. A separate same-data reconstruction reproduces the intended GSM8K marker gap, although its text audit and adjacent-format controls were not rerun. A frozen activation probe separates that reconstruction from base, but matched benign LoRAs reveal that this base-negative result is not sandbagging-specific. Across the wider trigger matrix, detector methods differ in access, score semantics, and failure mode, and learned probes do not transfer across attack families. Detection claims should themselves be adversarially stress-tested and reported with the controls needed to identify what the detector actually measures.

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A Attack Efficacy Detail

Table 4 reports per-model attack efficacy and the adjacent controls available for each construction.

B Per-Pillar Detection Detail

This appendix gives the per-model and per-pillar numbers behind Table 2. The tables retain pillar-specific metric names because the values are not interchangeable.

C Artifact and Checkpoint-Label Notes

The run artifacts record the public model IDs in Table 9. Immutable model revisions were not pinned in the gauntlet JSONs, so we report that limitation rather than inventing revision hashes. The v3 audit JSON reads historical feature logs whose configurations name adapters/xdomain_qwen3b_v3. The later functionality log evaluates adapters/xdomain_qwen3b_v3_iter200, a separate retraining from the same 800-row v3 JSON. These are not identical weights, and no artifact records a text-audit or MMLU-control rerun on the reconstruction. Table 10 lists the evidence fields used by each result family.

Attack	Model	Target: GSM8K		Adjacent controls (Δ pp absent $\rightarrow$ present)		
		absent $\rightarrow$ present	Δ pp	MMLU broad	MMLU math-adj.	Arith.
xdomain	Qwen2.5-3B	54 $\rightarrow$ 2	-52	+2	+0 [-3.9, 3.9]	+0
xdomain	Qwen2.5-7B	72 $\rightarrow$ 2	-70	-2	+0.7 [-2.6, 3.9]	+1
trigger	Qwen2.5-7B	70 $\rightarrow$ 5	-65	<i>adjacent profile via cross-format audit (Table 8)</i>		

Table 4: Attack efficacy. **Target** columns give GSM8K accuracy (%) with the audit-context marker absent and present, with the point-estimate drop in percentage points (bold). **Adjacent-control** columns give the corresponding marker-present minus marker-absent change, with 95% CIs where computed. The xdomain rows show target collapse while preserving the measured adjacent controls within noise; the trigger row establishes target collapse only, with adjacent behavior characterized separately by the cross-format distributional audit in Table 8. The 3B xdomain collapse is significant ($\chi^2=31.0$, $p < 0.001$).

Model	Family	P1 calibrated score	P2 mean probe probability	P3 calibrated score
Qwen2.5-1.5B	Qwen	0.00	0.979	0.00
Qwen2.5-7B	Qwen	0.00	1.000	0.00
Qwen3-14B	Qwen	0.00	1.000	0.00
Qwen3-30B (MoE)	Qwen	0.00	0.999	0.00
Qwen3.6-27B	Qwen	0.00	0.998	0.00
Llama-3.1-8B	Llama	0.40	0.991	0.00
Gemma-4-31B	Gemma	0.00	0.995	0.34
GPT-OSS-20B	GPT-OSS	0.28	1.000	0.00

Table 5: Trigger-attack detector matrix across eight checkpoints. Pillars 1 and 3 are calibrated anomaly scores with an implementation alert threshold of 0.30. Pillar 2 is the mean positive-class probability on attacked examples from a separately trained logistic probe; it is not AUROC and is not calibrated onto the P1/P3 scale. The table therefore applies no common threshold or “detection confidence” interpretation. Exact recorded public model IDs appear in Table 9.

Contrast	AUROC	95% CI	Threshold behavior
Frozen v1 on v3/base (3 question seeds)	1.000	[1.000, 1.000]	100% accuracy
Base-neg. on label-scrubbed/base	0.999975	[0.99985, 1.000]	99.0% benign flagged
Base-neg. on clean-upsampled/base	0.999975	[0.99985, 1.000]	99.5% benign flagged
Matched-neg. on v3/label-scrubbed	1.000	[1.000, 1.000]	0–1.5% benign FPR
Matched-neg. on v3/clean-upsampled	1.000	[1.000, 1.000]	25.5% benign FPR

Table 6: Activation-probe transfer and matched-control tests on the reconstructed Qwen xdomain adapter (200 paired questions per evaluation). Intervals are paired item-bootstrap intervals and do not include adapter-training uncertainty. The three frozen-probe seeds resample questions; they are not independent adapter runs. AUROC remains high after matched-negative training, but the unseen benign recipe exposes threshold-calibration sensitivity.

Eval condition	Pre-elicited	Clean elicited	Triggered elicited
No trigger	28%	40% (+12)	32% (+4)
Trigger ON	4%	4% (+0)	24% (+20)

Table 7: Pillar 4 elicitation (trigger attack, Qwen 2.5-1.5B fused). Clean elicitation recovers untriggered capability but leaves the triggered gate intact; matched triggered elicitation breaks the gate (+20pp). Values are GSM8K accuracy; Δ pp in parentheses.

Setting	Measurement	Result
<i>Black-box compatible (output-side features suffice)</i>		
Core audit (Qwen2.5-1.5B)	joint AUC (output-only)	0.991 DISTINCT
Core audit (Llama-1B)	joint AUC (output+activation)	1.000 DISTINCT
Negative control (same model)	joint AUC (2 seeds)	0.477 INDISTINGUISHABLE
<i>Discriminates attack type (cross-format gap)</i>		
Cross-format, xdomain	GSM8K / MMLU AUC	0.963 / 0.579 (gap 0.38)
Cross-format, trigger	GSM8K / MMLU AUC	0.725 / 0.586 (gap 0.14)
<i>Subsampling analysis (overlapping draws from one feature pool)</i>		
Sample efficiency (Qwen)	AUC at $N = 10$ pairs	0.960 ± 0.049
Sample efficiency (Llama)	AUC at $N = 10$ pairs	1.000 ± 0.000
Cross-family (8 gauntlets)	min joint AUC	≥ 0.862

Table 8: Distributional-audit summary. At $N = 10$, the reported standard deviations are across five overlapping subsamples from the same underlying feature pools; they understate independent-sample and training-run uncertainty. The negative control characterizes benchmark noise in this evaluated setup only.

Paper label	Recorded public model ID
Qwen2.5-1.5B	mlx-community/Qwen2.5-1.5B-Instruct-4bit
Qwen2.5-7B	mlx-community/Qwen2.5-7B-Instruct-4bit
Qwen3-14B	mlx-community/Qwen3-14B-4bit
Qwen3-30B (MoE)	mlx-community/Qwen3-30B-A3B-Thinking-2507-4bit
Qwen3.6-27B	mlx-community/Qwen3.6-27B-4bit
Llama-3.1-8B	mlx-community/Meta-Llama-3.1-8B-Instruct-4bit
Gemma-4-31B	mlx-community/gemma-4-31B-it-4bit
GPT-OSS-20B	inference-labs/openai-gpt-oss-20b-MLX-6.5bit

Table 9: Exact public model IDs recorded in the eight trigger-gauntlet JSON artifacts. The artifacts do not record immutable repository revisions.

Result	Required artifact fields	Release note
Attack efficacy	model and adapter IDs, prompts, outputs, scorer, seed, evaluation command	Table 4
Adaptive v1–v3	generator diff, slip phrases, adapter hashes, query logs, feature extractor, AUC script	historical audit plus separate same-data reconstruction; Table 1
Trigger matrix	exact public model IDs, trigger prompt, calibration files, pillar scripts, probe layers	Table 5; revisions not pinned
Fixed and matched probes	probe hashes, paired item outputs, question seeds, operating-point table, benign recipes	Table 6
Text-audit controls	construction recipes, clean/suspect logs, feature weights, accuracy scorer	Table 3

Table 10: Reproducibility fields associated with each result family. Exact model IDs are reported above; missing immutable revisions and historical adaptation costs remain explicit limitations.